Ad Optimization Agent

AI-Powered Budget Allocation for Maximum CTR

BUS4 118S | Section 02 | Group 5

The Problem

How do you split a fixed daily ad budget across multiple channels when each channel performs differently every day?

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Channels

Search, Social, Display

$540

Daily Budget

Fixed total, variable split

14 Days

Time Window

Feb 1–14, 2026

How the Agent Works

1. **Read Performance**
   Pull CTR, clicks, and spend from CSV for each channel

2. **Reallocate Budget**
   Shift 10% toward top CTR performer, reduce lowest

3. **Apply Guardrails**
   Cap changes at ±20%, enforce 10% floor, check timeouts

4. **Generate Rationale**
   Gemini explains the decision in 2 to 3 sentences

Key Design Choice

The LLM (Gemini) only writes explanations. All budget math is handled by deterministic Python code with hard-coded guardrails. This keeps decisions predictable and auditable.

Guardrails

±20%

Max Daily Change

No channel budget swings more than 20% in a single day

10%

Minimum Floor

Every channel gets at least 10% of total budget ($54)

2 Days

Floor Timeout

Forced +5% bump if a channel sits at minimum too long

$540

Normalization

Budgets always scale to sum to the fixed daily total

Results vs Baseline

0.0379

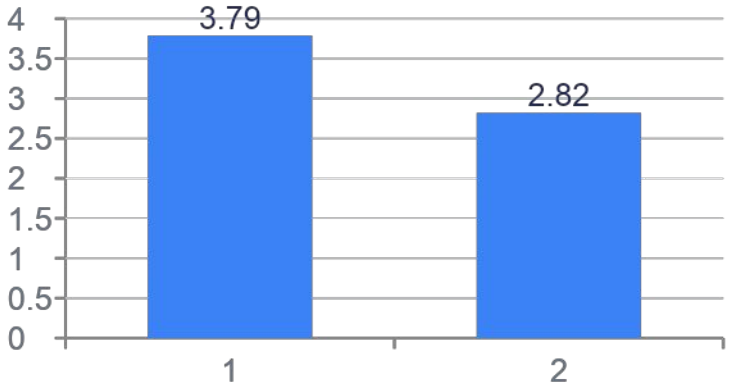
Agent Weighted CTR

0.0282

Baseline Weighted CTR

+34.57%

CTR Improvement



Key Findings

- Search dominated with ~4.8% CTR → budget grew to $429/day
- Display held at 10% floor (~$49–54/day)
- Social gradually reduced as middle performer

Channel Breakdown

Search

Avg CTR: 4.80%

Budget: $180 → $429

Top performer every day. Budget nearly tripled over the 14-day run.

Social

Avg CTR: 2.56%

Budget: $180 → $63

Consistent middle performer. Budget shrank as Search pulled ahead.

Display

Avg CTR: 1.10%

Budget: $180 → $49

Weakest CTR. Hit the 10% floor by Day 5 and stayed near it.

Scaling to Production

Infrastructure

- Deploy as Cloud Function or Airflow DAG
- Connect to live ad platform APIs
- Retry logic with exponential backoff
- Store decisions in BigQuery for audit

Controls & Monitoring

- Set daily API cost caps ($5–10/day)
- Add LangSmith / Datadog observability
- Alert on anomalous CTR swings
- Weekly performance review dashboard

Takeaways

1. Simple heuristic + guardrails delivered a 34.57% CTR improvement
2. The LLM adds explainability without controlling decisions
3. Minimum floors prevent data starvation on weaker channels
4. The framework is extensible to conversions, CPA, or multi-objective optimization